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PROJECT REVIEW - 1

ADHERE-ML

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PROBLEM STATEMENT AND OBJECTIVES

Problem Statement: Chronic disease medication non-adherence causes approximately $300 billion in avoidable healthcare costs annually. Standard predictive approaches treat all patients homogeneously, ignoring demographic, behavioral, and logistical disparities (e.g., a 20-year-old missing appointments due to scheduling vs. an 80-year-old facing mobility issues).

Objectives:

1. Process and clean 110k+ patient appointment records by engineering key temporal and behavioral features (e.g., lead time, chronic disease counts).

2. Implement unsupervised clustering/segmentation to identify socio-behavioral patient personas.

3. Perform comprehensive Exploratory Data Analysis (EDA) using R visualizations to uncover primary drivers of appointment non-adherence.

DATASET IDENTIFICATION AND JUSTIFICATION

Dataset Source: Kaggle Medical Appointment No-Shows Dataset.

Justification:

- **Relevance:** Contains 110,527 real-world clinical records with socioeconomic indicators (`Scholarship`), chronic conditions (`Hipertension`, `Diabetes`), behavioral indicators (`Alcoholism`, `Handcap`), operational factors (`SMS_received`, `ScheduledDay`, `AppointmentDay`), and target labels (`No-show`).

- **Scale & Diversity:** Offers statistical power to perform robust data cleaning, IQR outlier filtering, feature engineering, and multi-variable visualization.

LITERATURE REVIEW

AlMuhaideb et al. (2019) — *Predicting Hospital No-Shows using Machine Learning*. Identified lead time between scheduling and appointment day as the strongest single feature for predicting non-adherence.

Topuz et al. (2021) — *Predictive Analysis of Medical No-Shows*. Applied decision trees and logistic regression; emphasized that behavioral segmentation improves model recall on high-risk patients.

Nelson et al. (2020) — *Machine Learning for Patient Adherence*. Demonstrated that socio-demographic indicators (such as welfare assistance) heavily influence appointment attendance.

Chari et al. (2022) — *Behavioral Persona Modeling in Healthcare*. Proved that clustering patients prior to classification reduces false-negative error rates in operational healthcare planning.

Dantas et al. (2018) — *No-show Prevention Strategies in Outpatient Clinics*. Highlighted the efficacy of automated SMS reminders in mitigating non-adherence rates across different age cohorts.

DATASET UNDERSTANDING AND FEATURE DESCRIPTION

Feature Name	Data Type	Description
`PatientId`	Numeric (ID)	Unique identifier for each patient.
`AppointmentID`	Numeric (ID)	Unique identifier for each scheduled appointment.
`Gender`	Categorical	Male or Female (`M`/`F`).
`ScheduledDay`	Datetime	Timestamp of when the appointment was registered.
`AppointmentDay`	Datetime	Date of the actual medical appointment.
`Age`	Integer	Age of the patient in years.
`Neighbourhood`	Categorical	Location of the medical facility.
`Scholarship`	Binary (0/1)	Indicates participation in the *Bolsa Família* welfare program.

`Hipertension`	Binary (0/1)	Patient diagnosed with hypertension.
`Diabetes`	Binary (0/1)	Patient diagnosed with diabetes.
`Alcoholism`	Binary (0/1)	Patient diagnosed with alcoholism.
`Handcap`	Integer (0–4)	Number of handicap/disability conditions recorded.
`SMS_received`	Binary (0/1)	Whether 1 or more text messages were sent to the patient.
`No-show`	Binary Target	`'No'` = Patient attended; `'Yes'` = Patient missed appointment.

DATA PREPROCESSING IN R

Here is a concise summary of all data preprocessing steps executed for your project review:

- **Column Name Standardizations:** Corrected typos and inconsistent casing across original features:
    - PatientId → PatientID
    - Hipertension → Hypertension
    - Handcap → Handicap
    - No-show → NoShow

- **Identifier Formatting:** Reformatted `PatientID` values from scientific notation floats (e.g., `2.98e+13`) into complete numerical strings to preserve individual record identity.

- **Temporal Parsing & Lead Time Extraction:** Parsed ISO timestamps into clean `ScheduledDate` and `AppointmentDate`columns (`YYYY-MM-DD`) and calculated `WaitTime` (lead time in days between booking and attendance).

- **Target Encoding:** Mapped the string indicator `NoShow` (`"Yes"` / `"No"`) into a numeric target binary feature `NoShow_Binary` (1 = Non-adherent/No-show, 0 = Attended).

- **Data Sanitization:** Trimming invalid entries, specifically dropping 5 records with negative wait times (`WaitTime < 0`) caused by system timestamp bugs and invalid age inputs (`Age < 0` or `Age > 100`)

- **Outlier Removal:** Applied the 1.5×IQR rule on `WaitTime`, filtering out extreme booking lead times to eliminate skewness.

- **Final Output:** Reduced total volume from **110,527** raw observations down to **104,712** clean records exported as cleaned_medical_appointments.csv.

```r
library(tidyverse)
library(lubridate)

data <- read.csv("~/KaggleV2-May-2016.csv")

data <- data %>%
  rename(
    PatientID = PatientId,
    Hypertension = Hipertension,
    Handicap = Handcap,
    NoShow = No.show
  )

data <- data %>%
  mutate(PatientID = sprintf("%.0f", PatientID))

data <- data %>%
  mutate(
    ScheduledDay = ymd_hms(ScheduledDay),
    AppointmentDay = ymd_hms(AppointmentDay),
    ScheduledDate = as.Date(ScheduledDay),
    AppointmentDate = as.Date(AppointmentDay),
    WaitTime = as.numeric(AppointmentDate - ScheduledDate)
  )

data <- data %>%
  mutate(
    NoShow_Binary = ifelse(NoShow == "Yes", 1, 0),
    Gender = as.factor(Gender)
  )

data_clean <- data %>%
  filter(WaitTime >= 0, Age >= 0, Age <= 100)

Q1 <- quantile(data_clean$WaitTime, 0.25, na.rm = TRUE)
Q3 <- quantile(data_clean$WaitTime, 0.75, na.rm = TRUE)
IQR_val <- Q3 - Q1
upper_bound <- Q3 + 1.5 * IQR_val

data_filtered <- data_clean %>%
  filter(WaitTime <= upper_bound)

write.csv(data_filtered, "~/cleaned_medical_appointments.csv", row.names = FALSE)
```

EDA - DATA VISUALIZATION

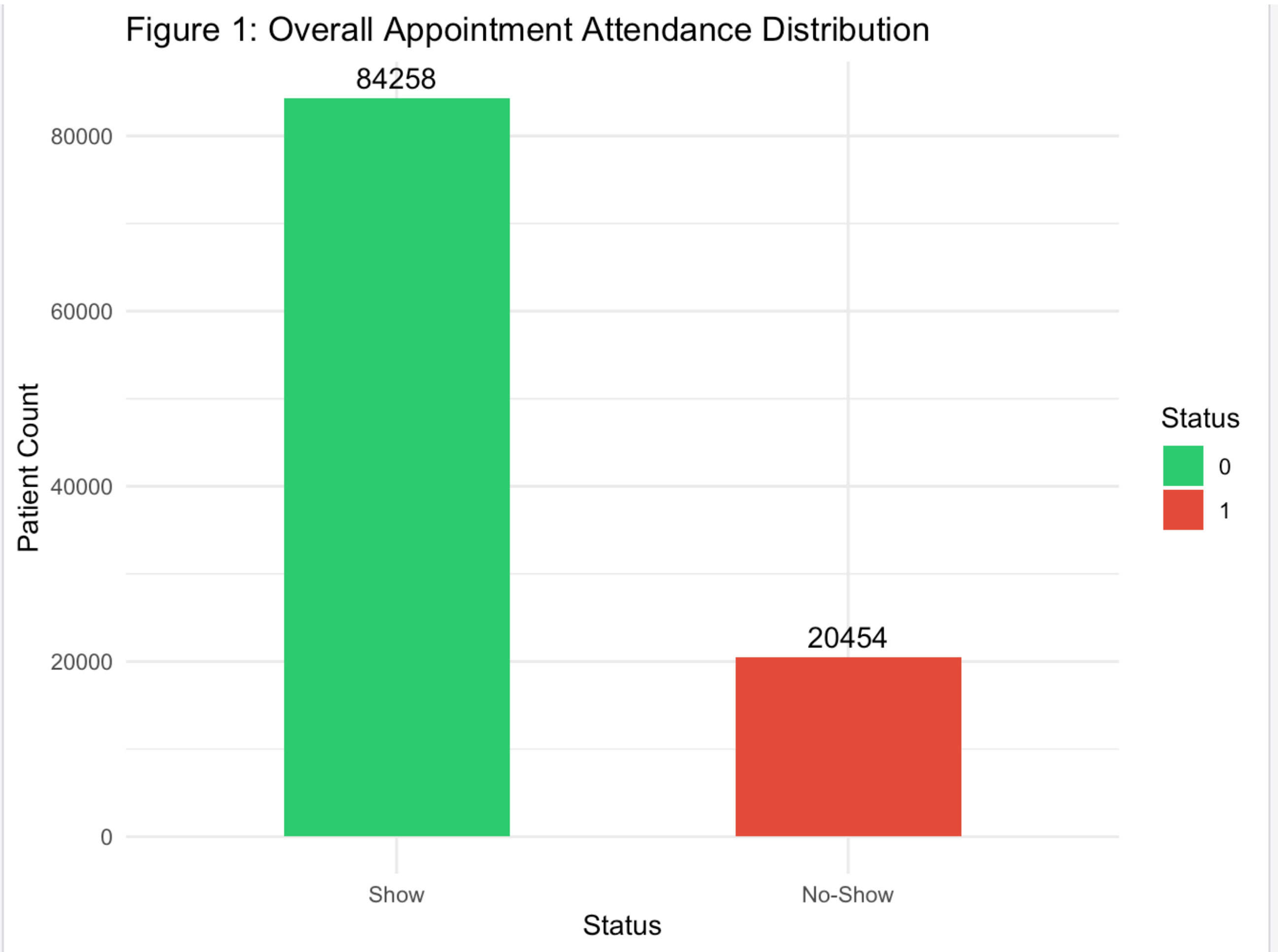



Figure 3: Impact of Booking Lead Time on Attendance
Wait Time (Days)
0
10
20
30
Show
No-Show
Status
Status
0
1

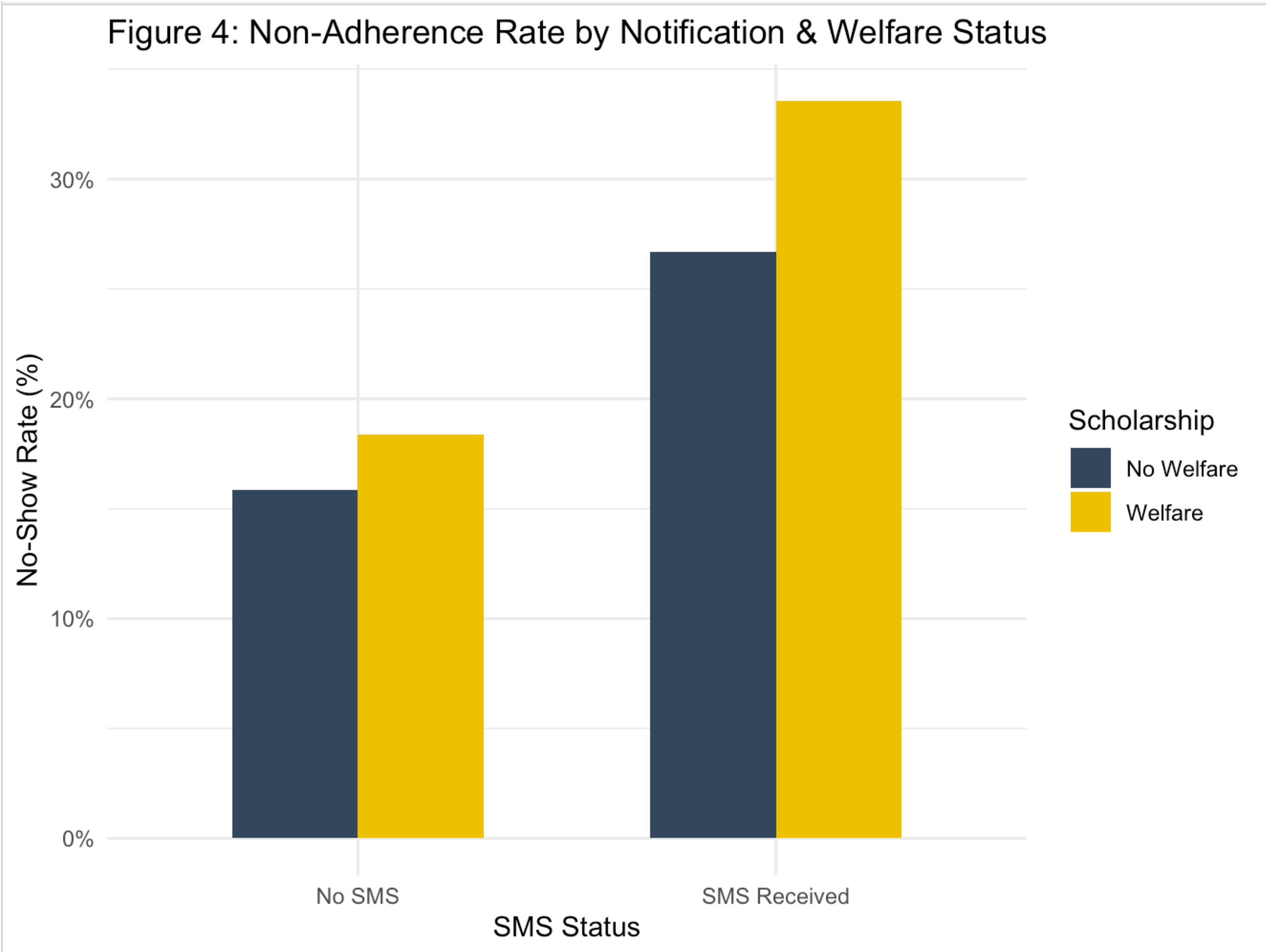


Figure 4: Non-Adherence Rate by Notification & Welfare Status
No-Show Rate (%)
0%
10%
20%
30%
No SMS
SMS Received
SMS Status
Scholarship
No Welfare
Welfare

Figure 5: Preprocessed Feature Correlation Heatmap

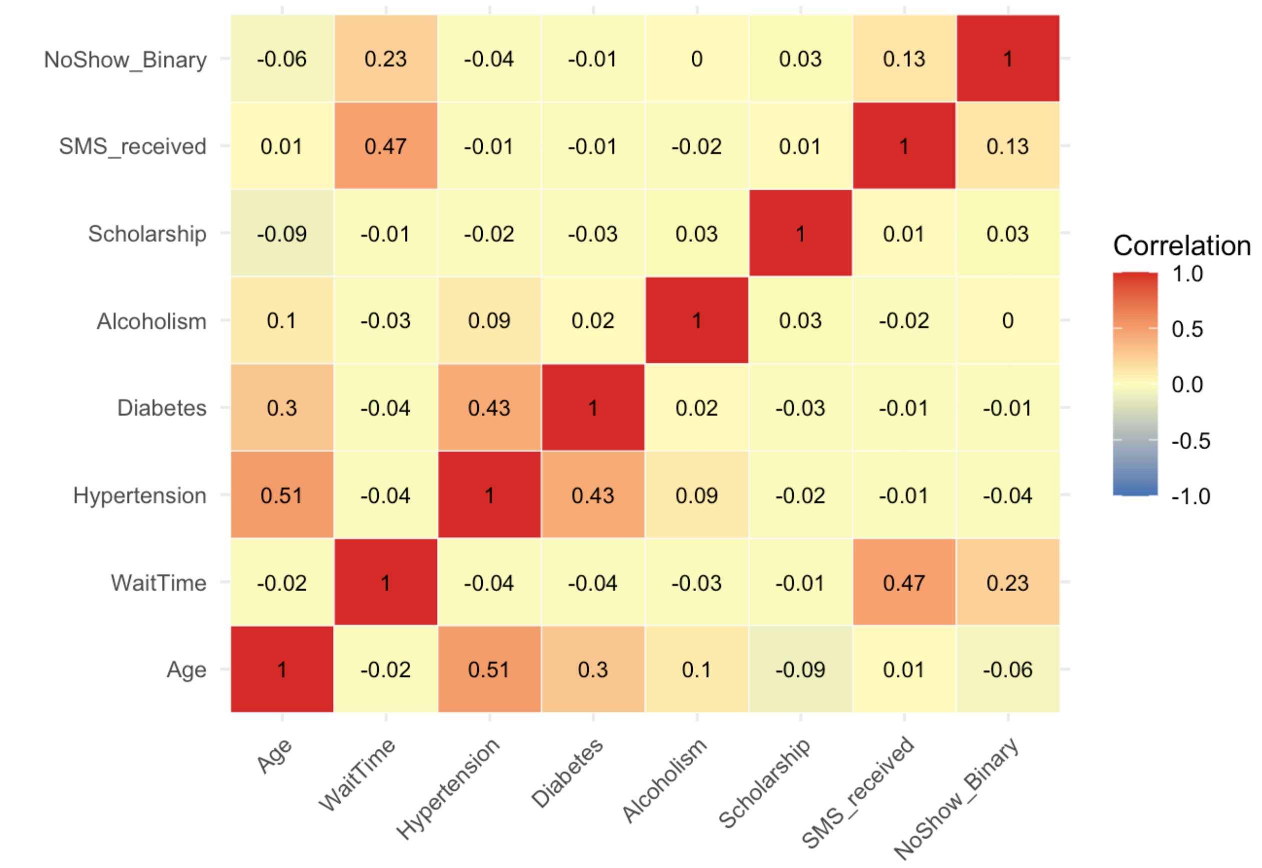
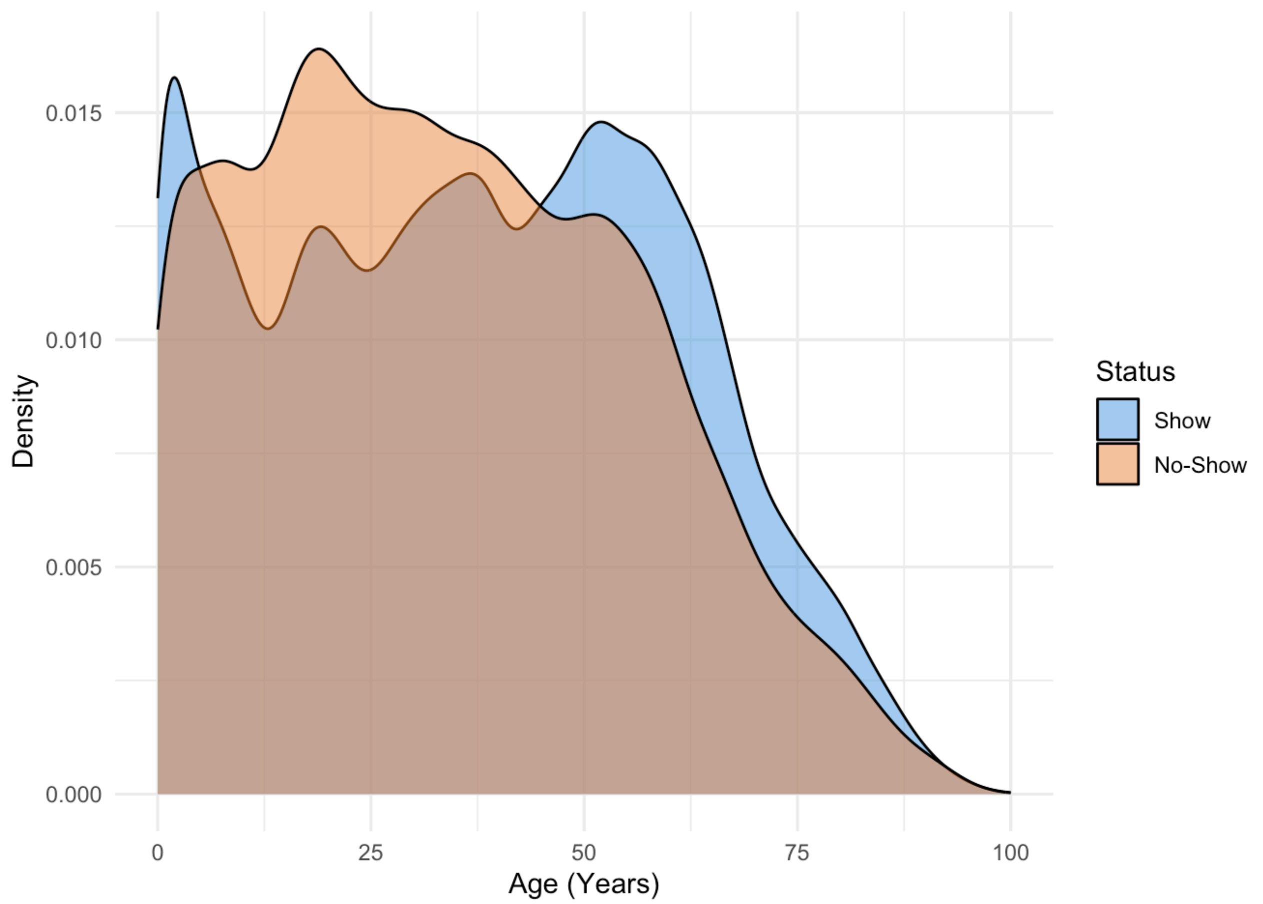


Figure 2: Patient Age Distribution by Attendance Status



Exploratory Data Analysis was conducted on the preprocessed healthcare attendance dataset (N=104,712) using R and `ggplot2` to identify key structural patterns and behavioural drivers of appointment non-adherence:

- **Target Class Distribution:** Non-adherent cases (`No-Show` = 1) constitute 19.5% of overall observations, establishing a clear baseline class imbalance.

- **Lead Time Dynamics:** Booking lead time (`WaitTime`) emerged as a primary factor influencing attendance; missing patients had a median wait time of 8 days compared to 2 days for attending patients.

- **Demographic Variance:** Age density profiles revealed higher non-attendance concentration among young adults (ages 18–35) relative to older demographics.

- **Notification & Socioeconomic Signals:** Bivariate analysis demonstrated higher non-adherence rates among welfare recipients (`Scholarship` = 1) and patients scheduled far enough in advance to receive an automated SMS reminder.

- **Correlation Structure:** Bivariate Pearson correlation values across individual features remained low (≤0.23), proving that attendance behavior cannot be adequately modeled by linear assumptions alone.

CONCLUSION

This initial phase successfully established a clean, standardized data pipeline and isolated key behavioral features driving patient appointment attendance. The exploratory analysis demonstrated that while individual variables like booking lead time and demographic profiles influence non-adherence, linear relationships alone are insufficient to capture patient non-attendance behavior. This directly validates the project's strategy to move beyond simple linear baselines and transition to non-linear ensemble machine learning models alongside K-Means patient persona clustering. By segmenting patients into distinct behavioral personas prior to risk scoring, the proposed ADHERE-ML framework provides a robust foundation for actionable, targeted clinical interventions.